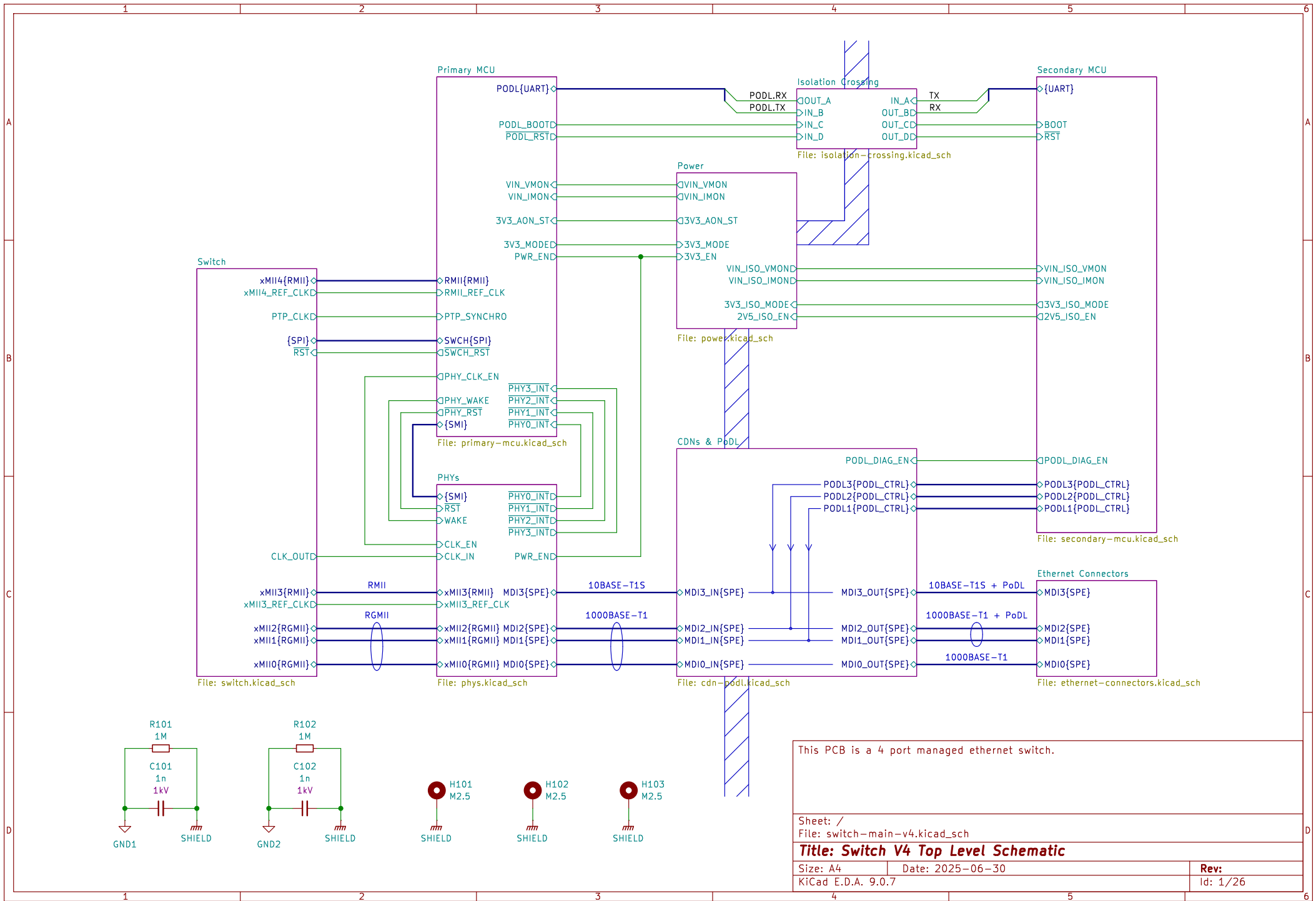
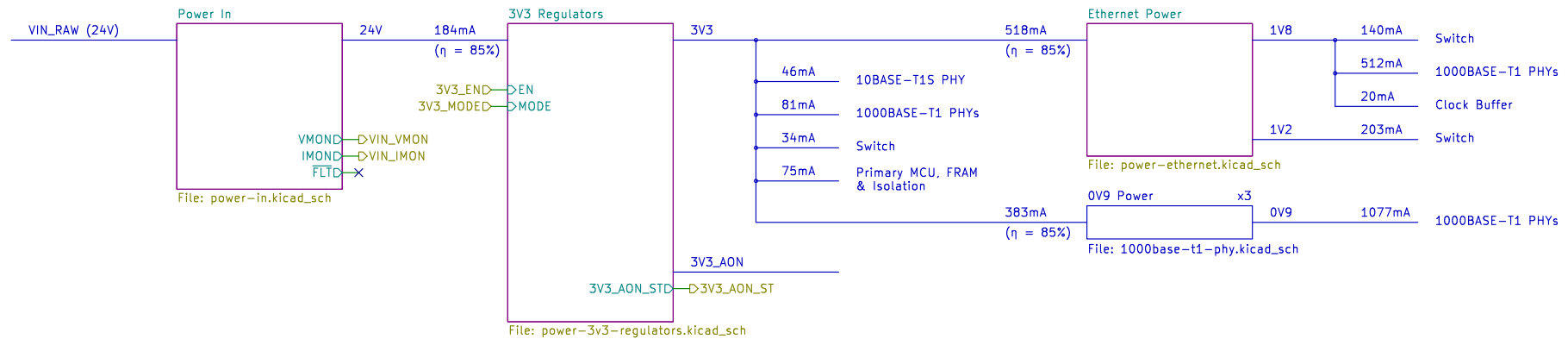


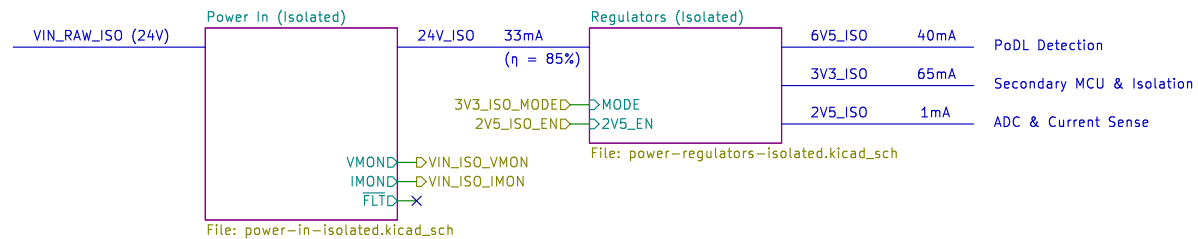


PHY Side Power



Isolation Barrier

Connector Side Power



Block diagram of the power supply scheme for both sides of the isolation barrier.

Sheet: /Power/
File: power.kicad_sch

Title: Power

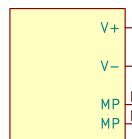
Size: A4	Date: 2025-06-30
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J301
MKXS1-1 Power Input



ESD
Protection

Surge
Protection

VIN_RAW

D301
SZESD7241N2T5G



U301
TVS3301DRBR

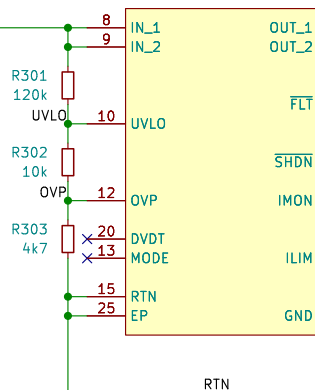
C301
100n
100V

GND1

UVLO = 10.9V
OVP = 34.1V

eFuse

U302
TPS26600RHFR



$VMON = VIN / 18.857$
The 150mΩ eFuse
resistance should be
taken into account when
calculating input power.

+VIN

R307
100k

VMON

R309
5k6

GND1

$I_{OUT} = VMON / 7.84$

$R_{ILIM} = 47k\Omega \therefore$
 $I_{LIM} = 255mA$

TPS26600 has auto-retry
when a fault occurs.

Power input and protection for the PHY side of the PCB.

Sheet: /Power/Power In/
File: power-in.kicad_sch

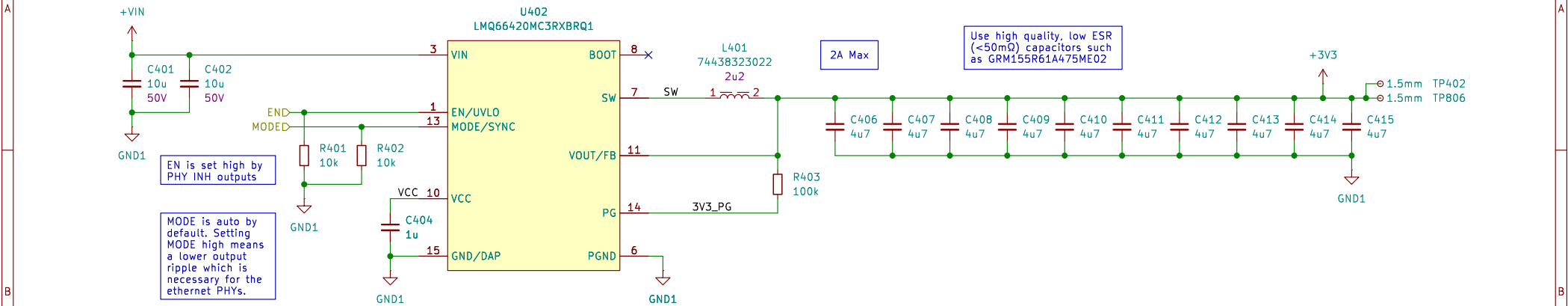
Title: Power In (PHY Side)

Size: A4 Date: 2025-06-30

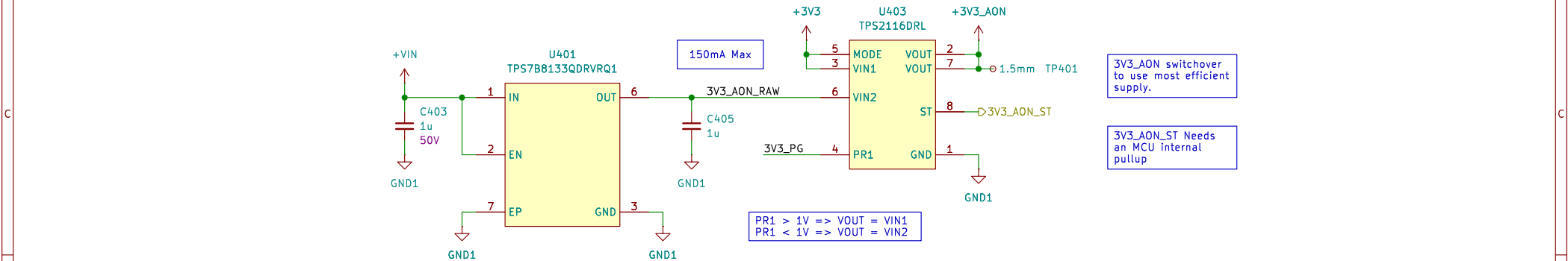
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DATE: 11/11/2015 11:00 AM



3V3 and 3V3_A0N (always on) voltage regulators.

Sheet: /Power/3V3 Regulators/
File: power-3v3-regulators.kicad_sch

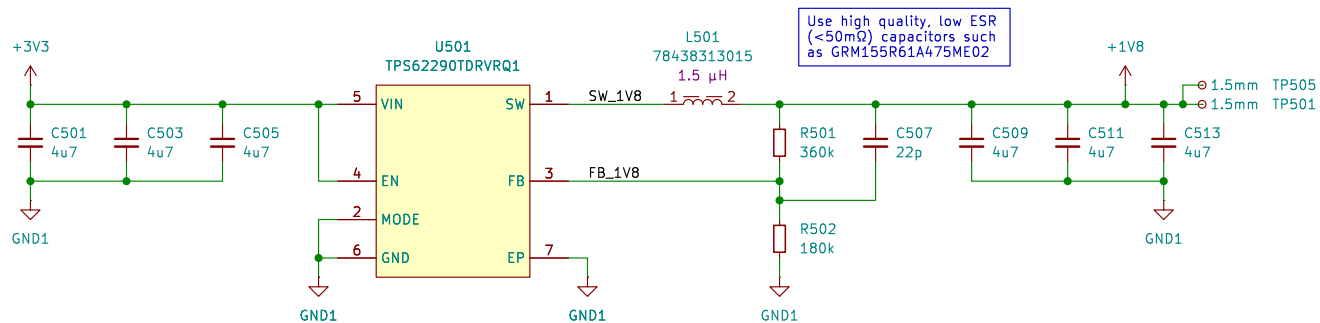
Title: 3V3 Regulators (PHY Side)

Size: A4	Date: 2025-06-30
KiCad E.D.A. 9.0.7	

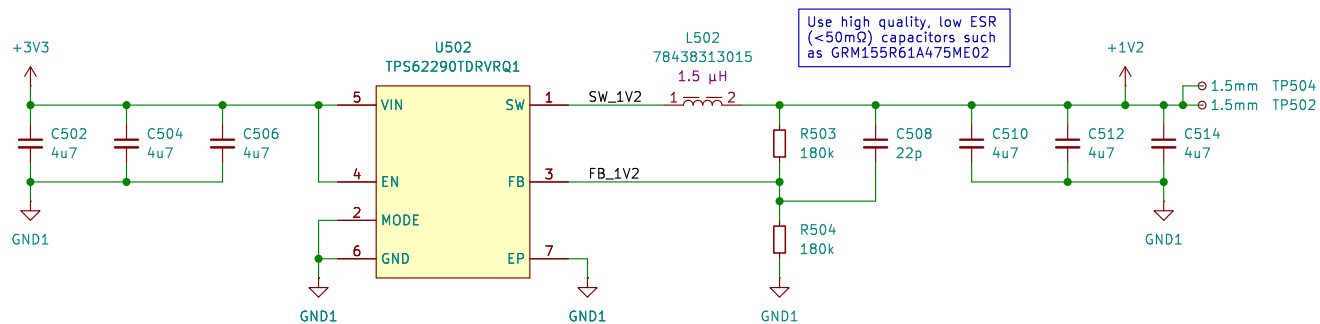
Date: 2025-06-30

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1V8 Regulator



1V2 Regulator



1V8 and 1V2 SMPS for ethernet switch and PHYs.

Sheet: /Power/Ethernet Power/
File: power-ethernet.kicad_sch

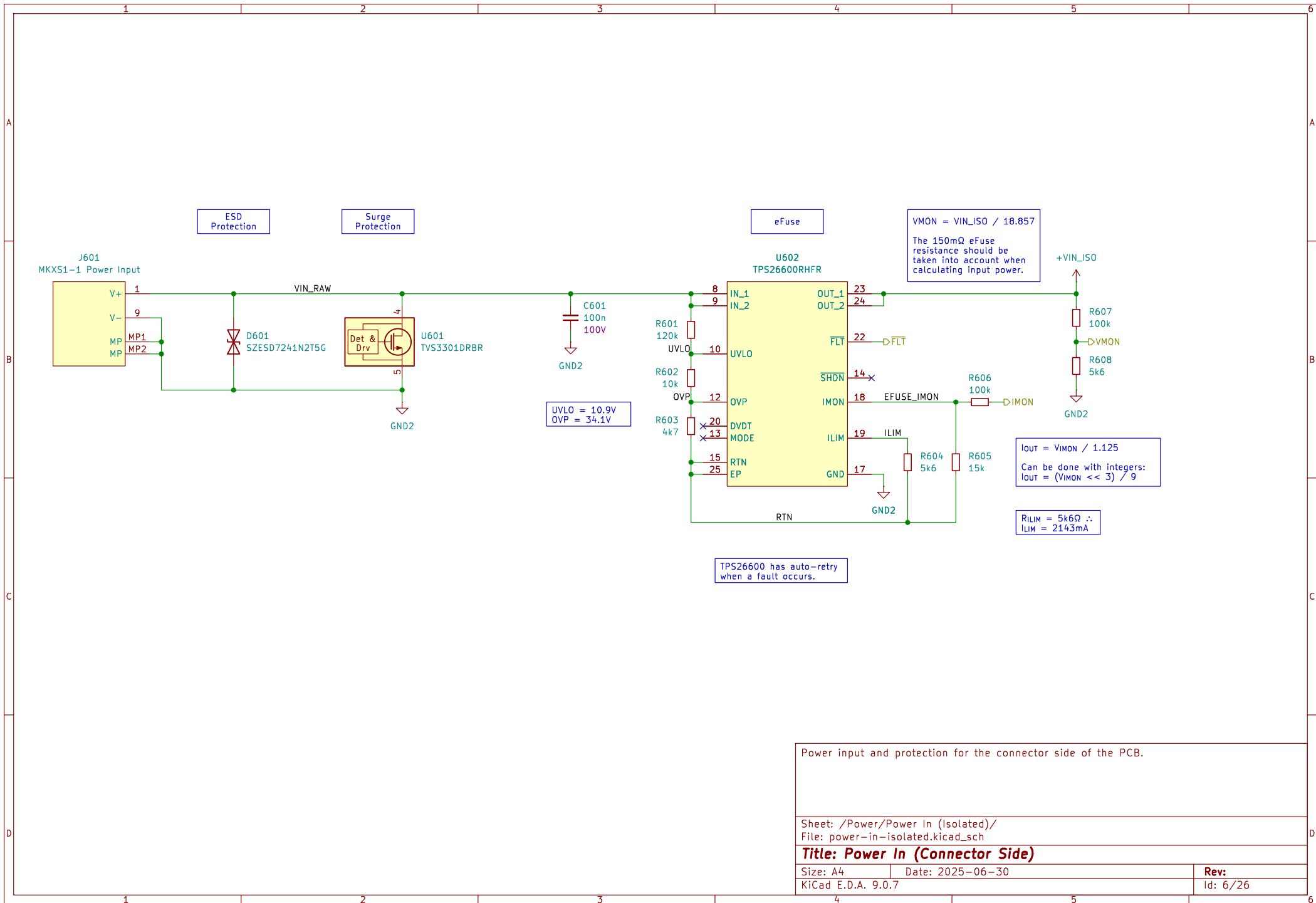
Title: Ethernet Power (PHY Side)

Size: A4 Date: 2025-06-30

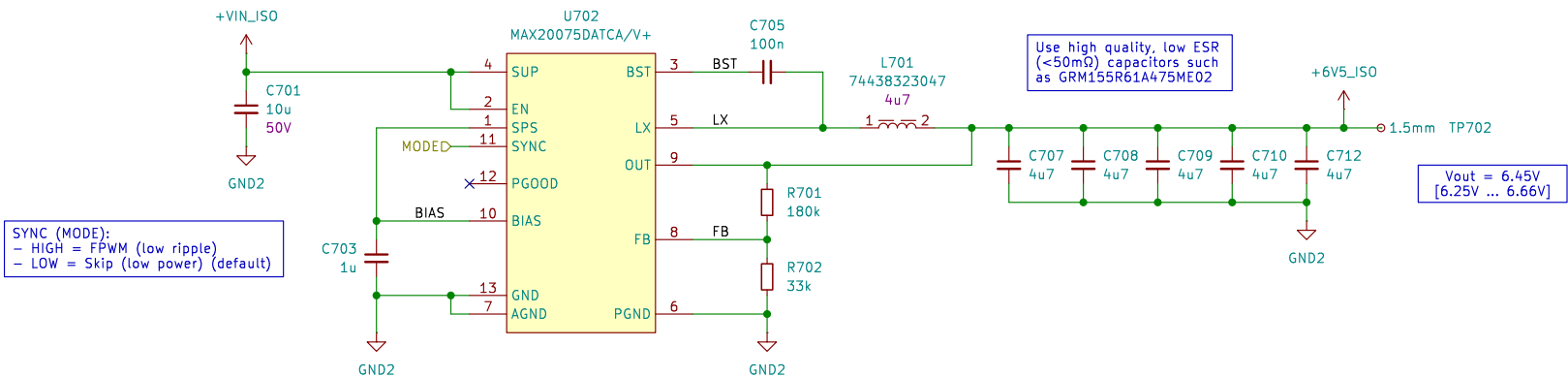
KiCad E.D.A. 9.0.7

Rev:

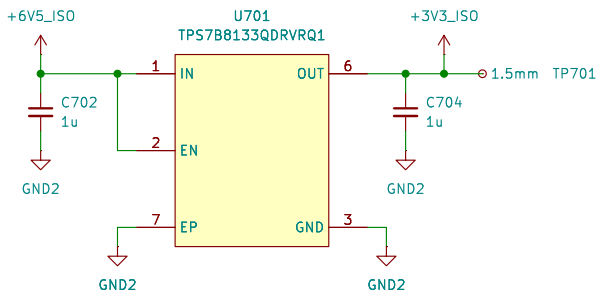
Id: 5/26



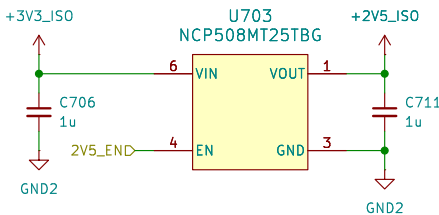
6V5 Regulator



3V3 Regulator



2V5 Regulator



6V5, 3V3 and 2V5 regulators for the connector side of the PCB.

Sheet: /Power/Regulators (Isolated)/
File: power-regulators-isolated.kicad_sch

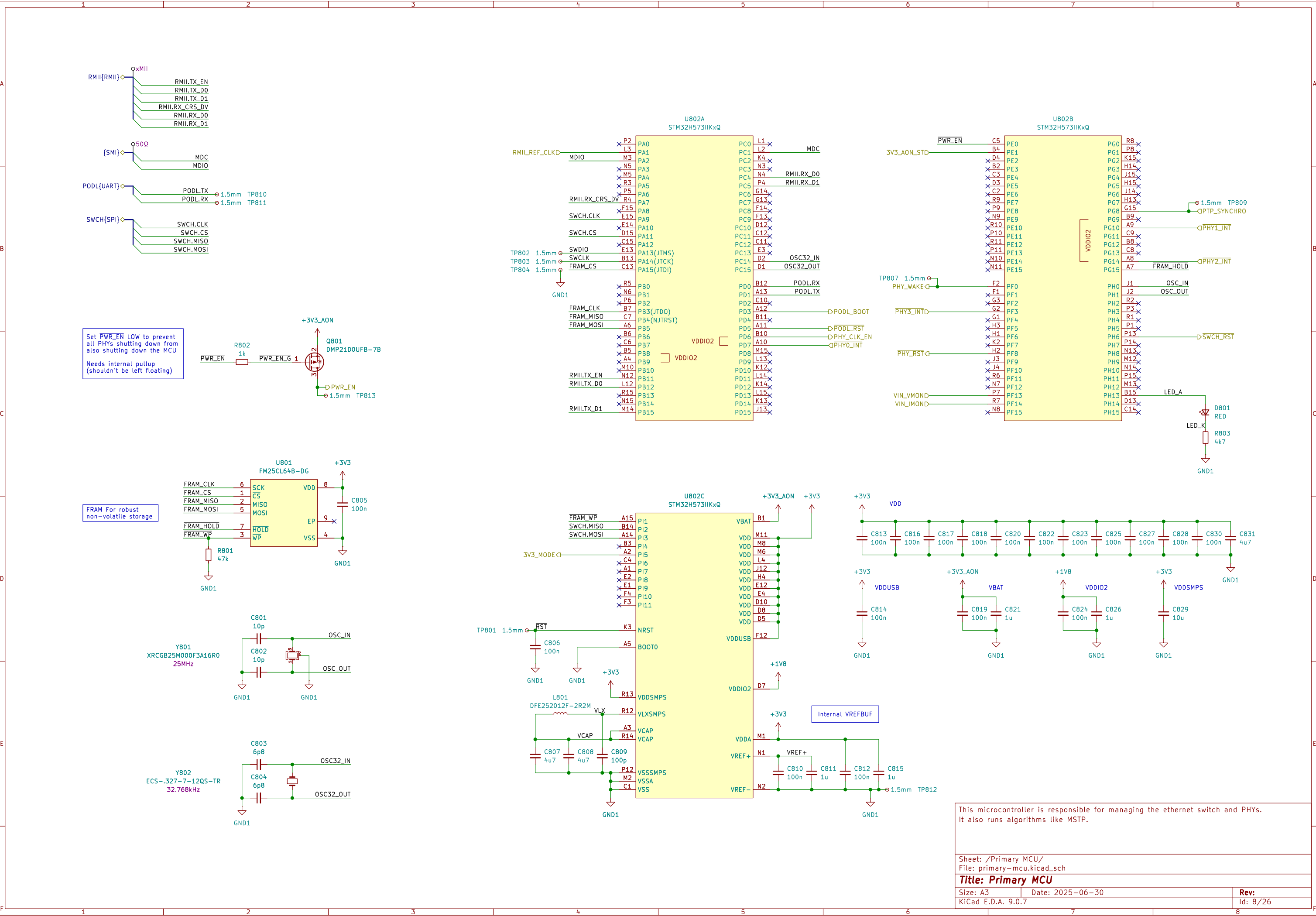
Title: Regulators (Connector Side)

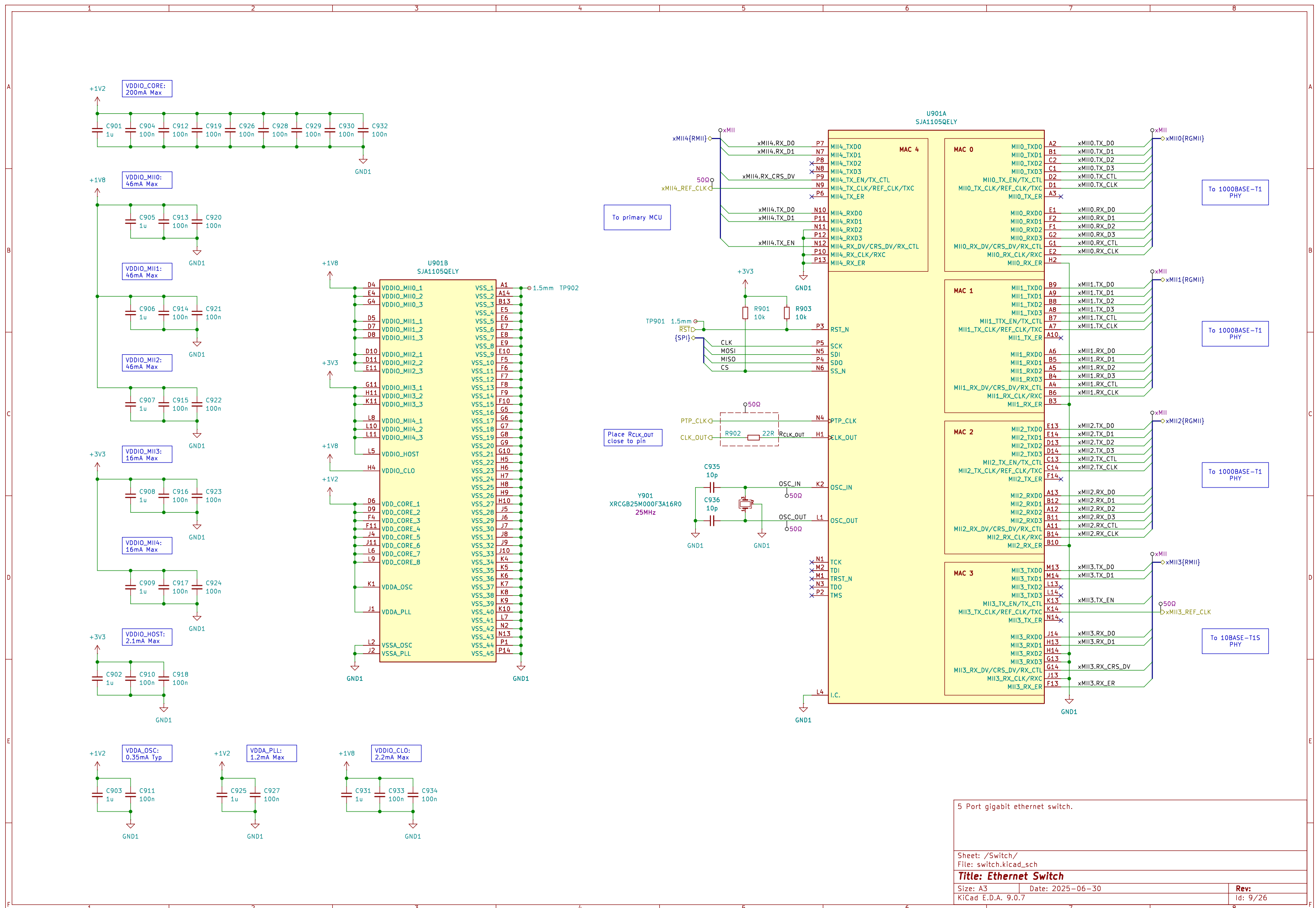
Size: A4 Date: 2025-06-30

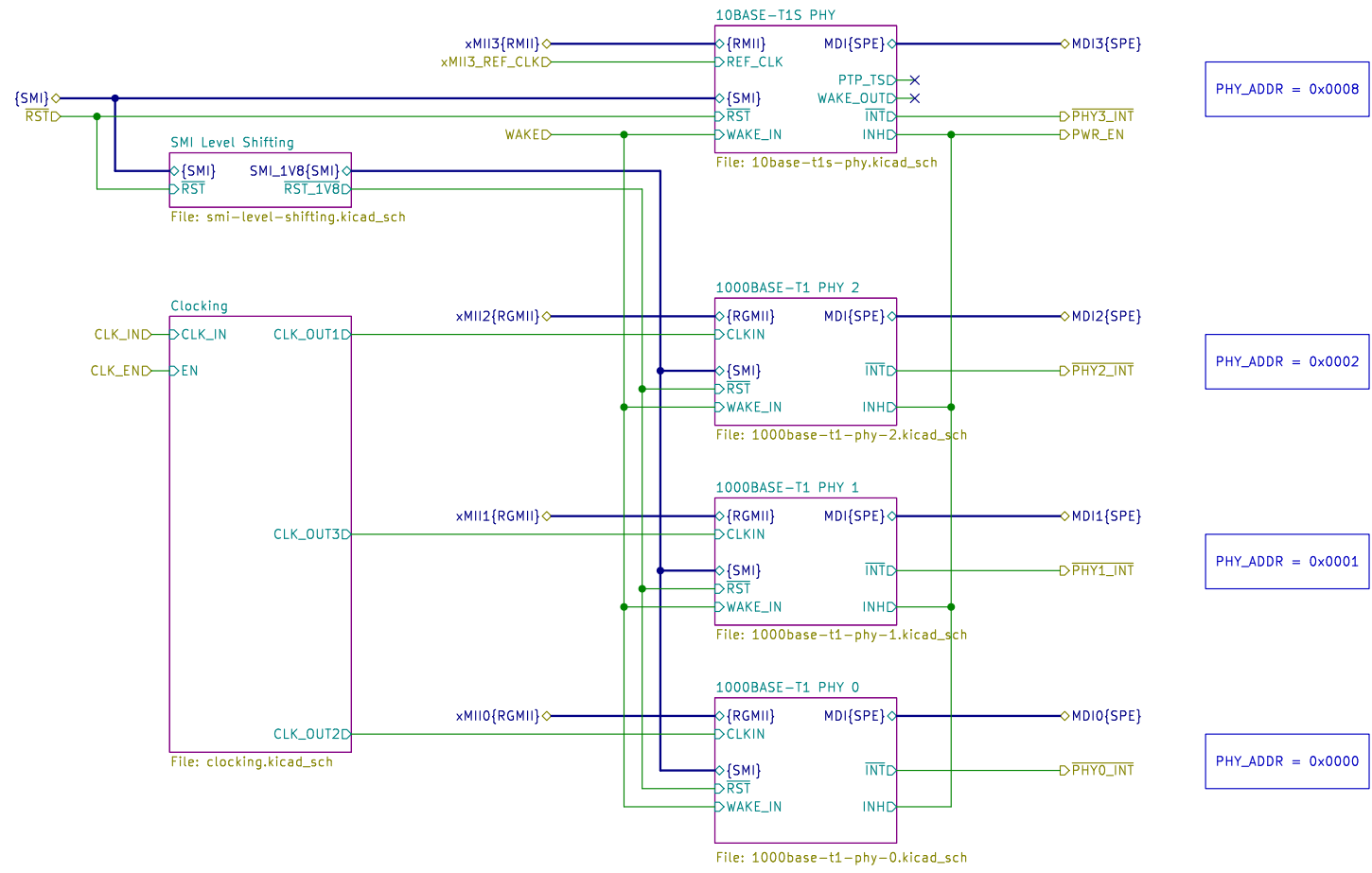
KiCad E.D.A. 9.0.7

Rev:

Id: 7/26







Ethernet PHYs and supporting circuitry.

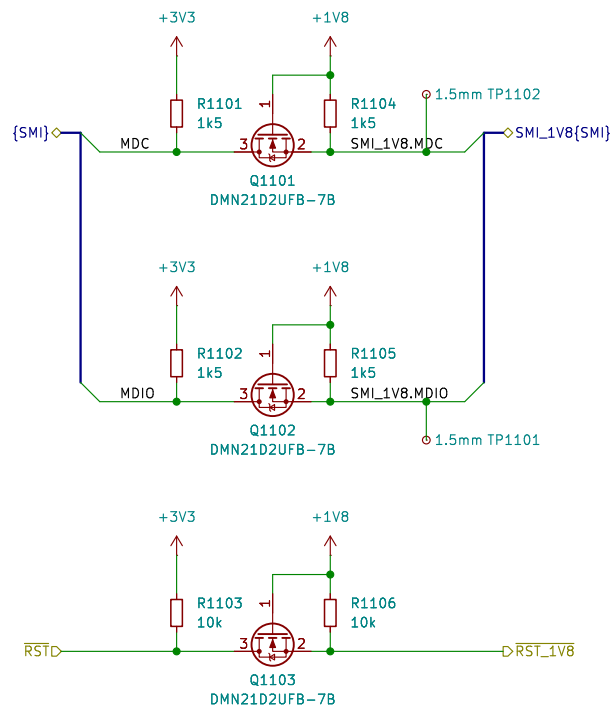
Sheet: /PHYs/
File: phys.kicad_sch

Title: PHYs

Size: A4 Date: 2025-06-30

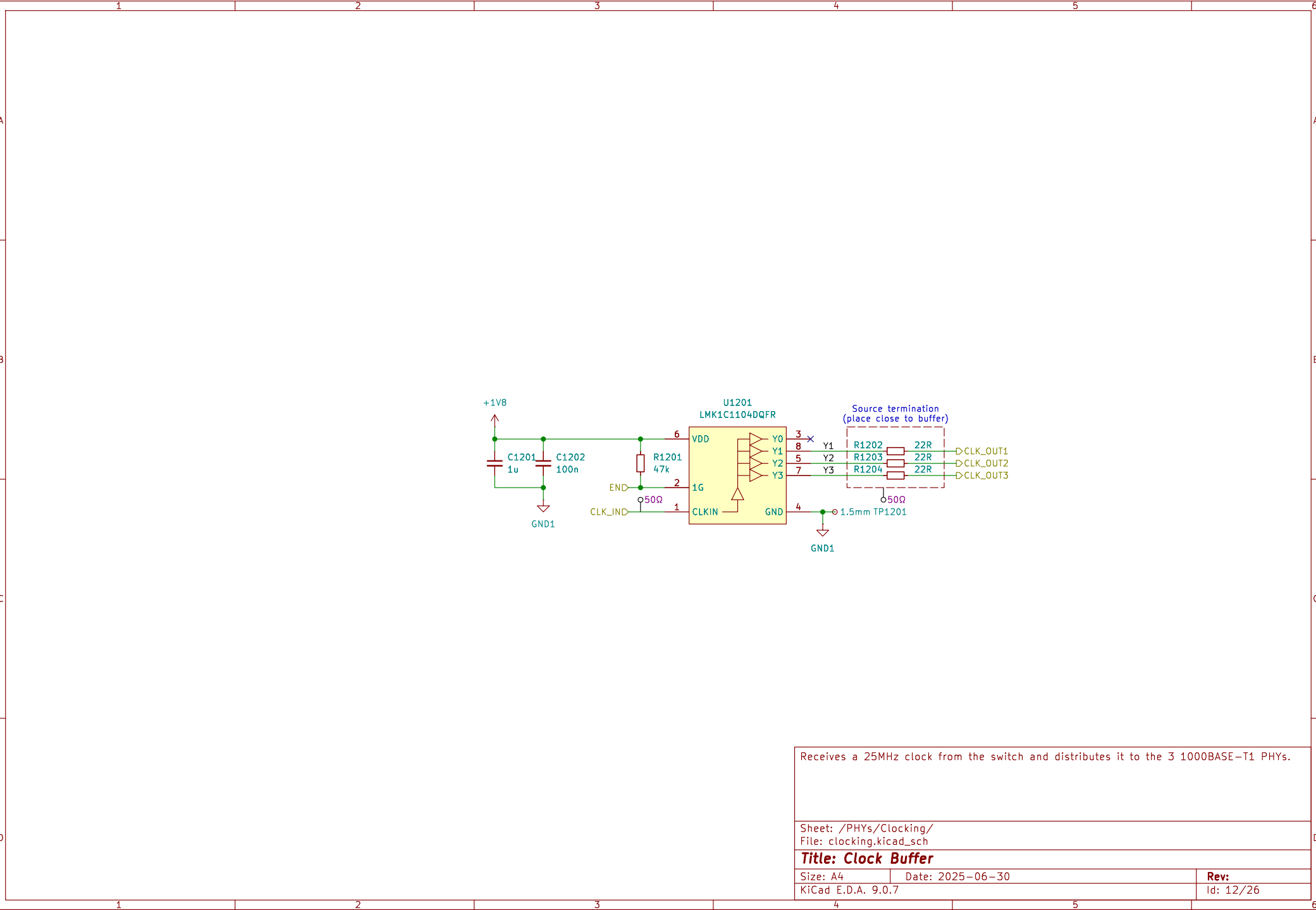
KiCad E.D.A. 9.0.7

Rev:
Id: 10/26



The 1000BASE-T1 PHYs operate at a 1V8 logic level (to save RGMII power), but the SMI and RST pins operate at a 3V3 logic level. Level shifting is used to allow communication.

Converts 3.3V PHY management signals to 1.8V.		
Sheet: /PHYs/SMI Level Shifting/ File: smi-level-shifting.kicad_sch		
Title: SMI Level Shifting		
Size: A4	Date: 2025-06-30	Rev:
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Receives a 25MHz clock from the switch and distributes it to the 3 1000BASE-T1 PHYs.

Sheet: /PHYs/Clocking/
File: clocking.kicad_sch

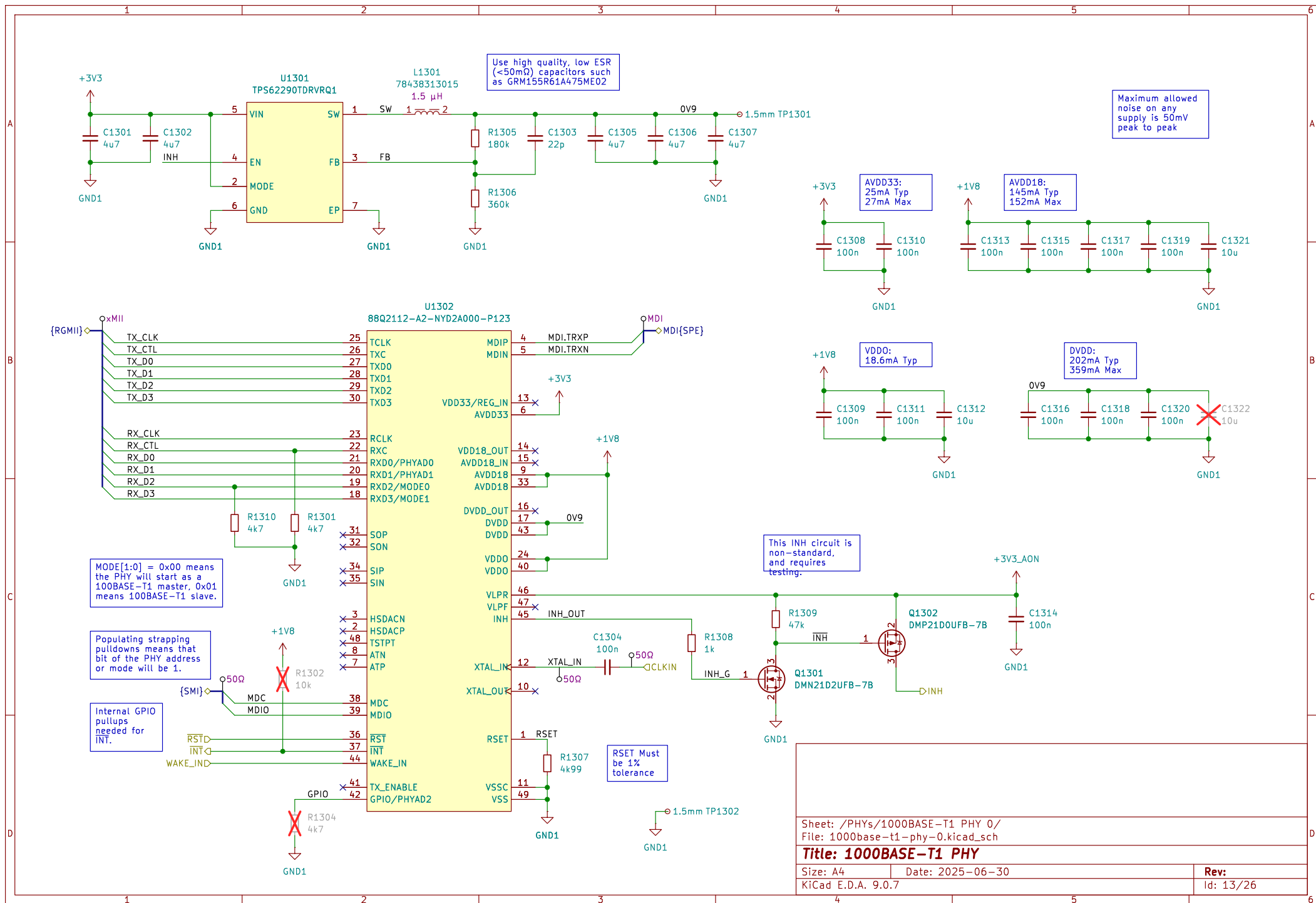
Title: Clock Buffer

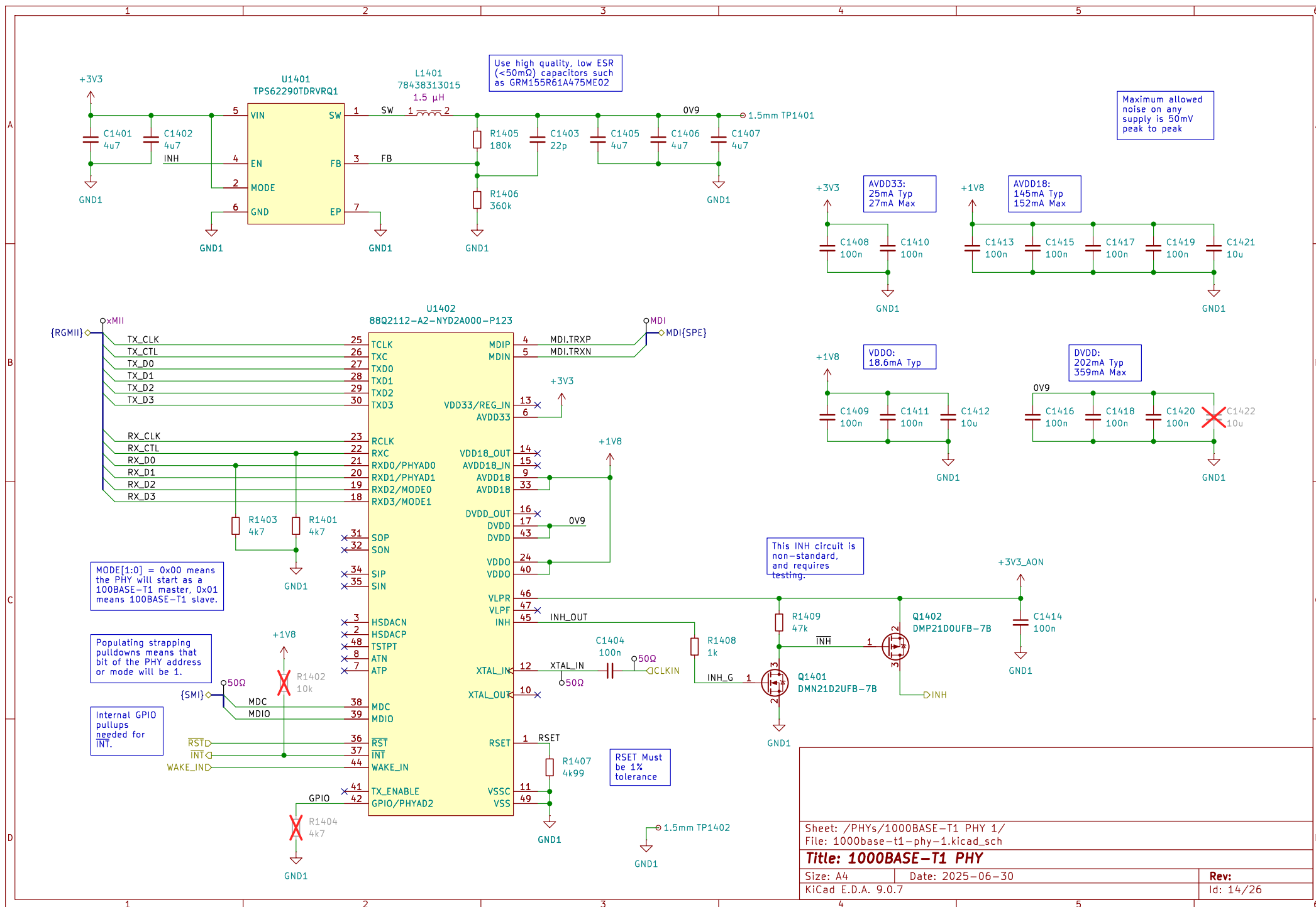
Size: A4 Date: 2025-06-30

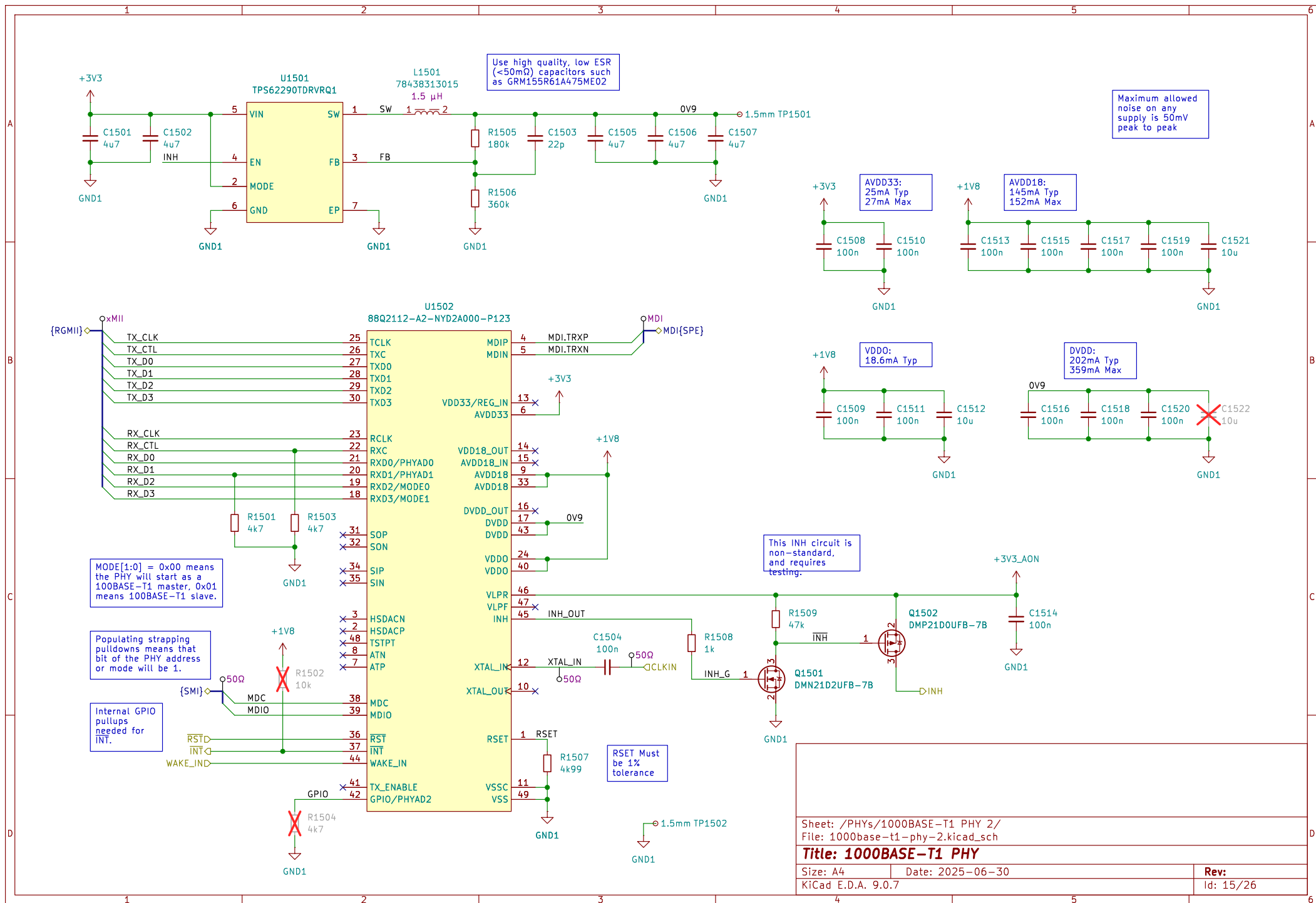
KiCad E.D.A. 9.0.7

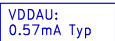
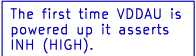
Rev:

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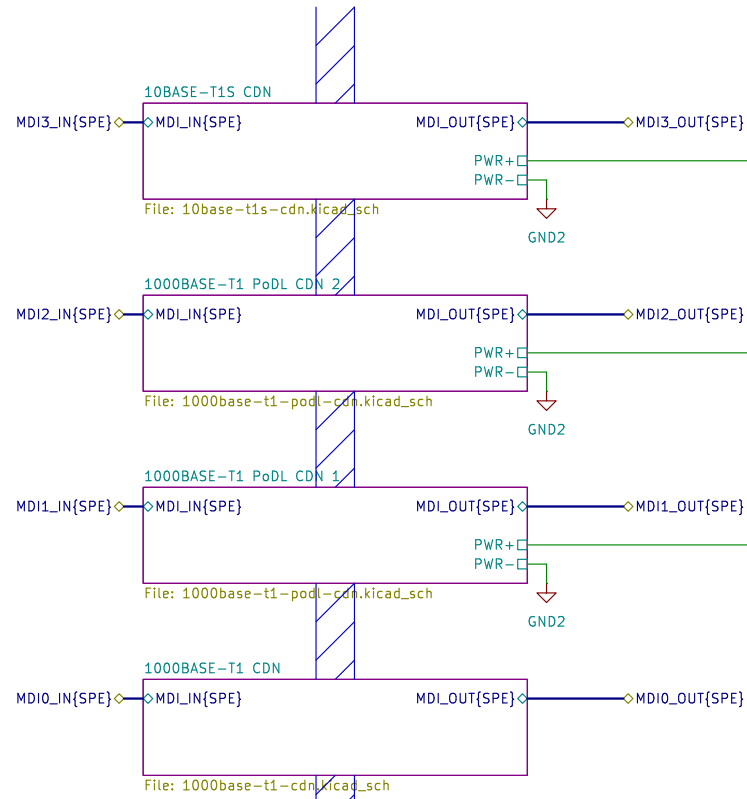




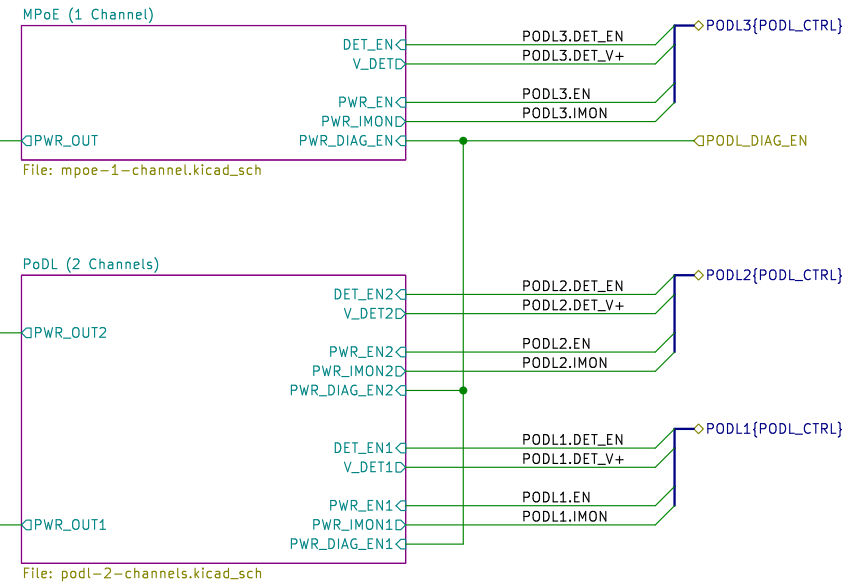


Id: 16/26

Coupling-Decoupling Networks (CDNs)



Power Injection



Coupling-decoupling networks (CDNs) and power over datalines (PoDL).

Sheet: /CDNs & PoDL/
File: cdn-podl.kicad_sch

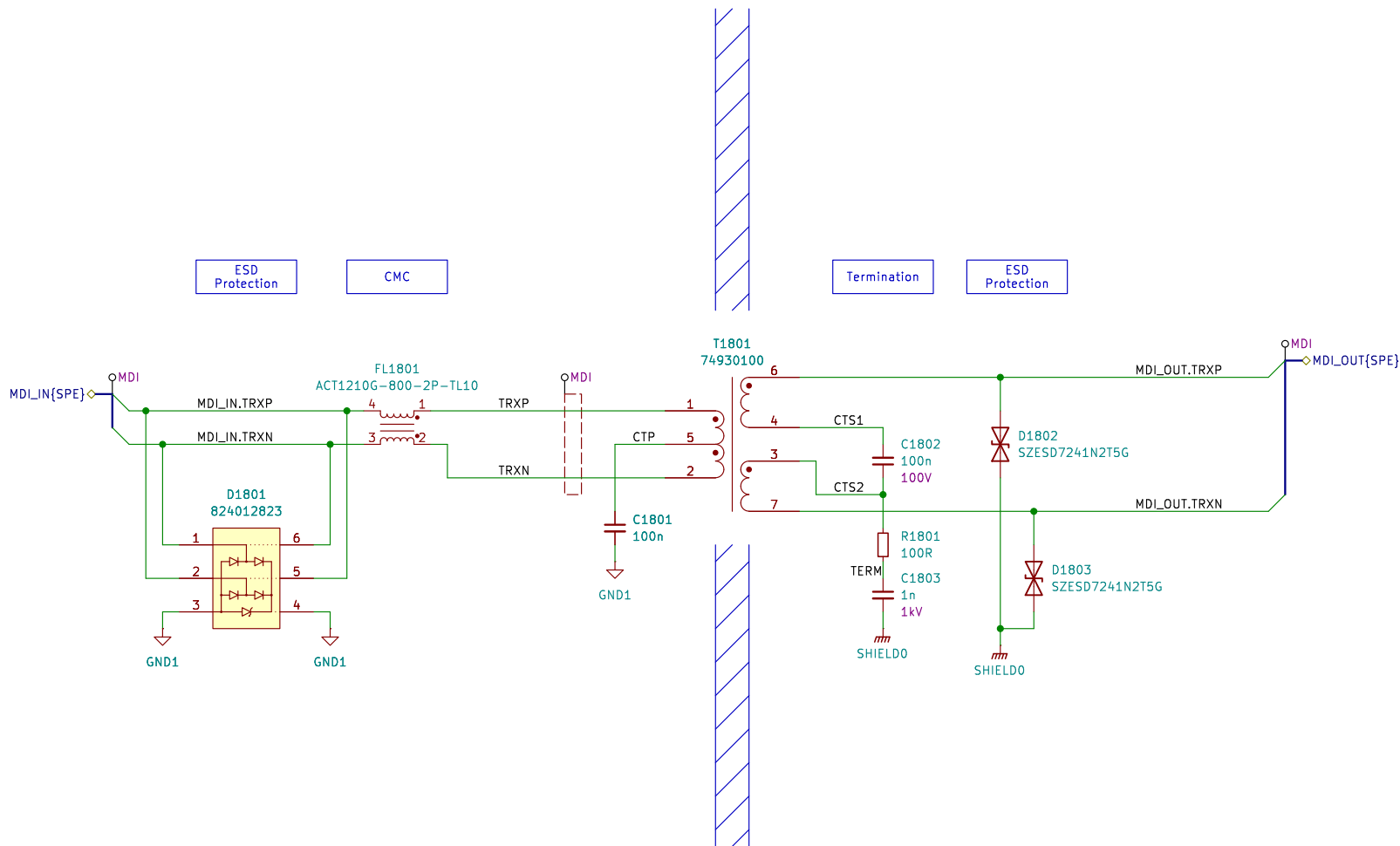
Title: CDNs and PoDL

Size: A4 Date: 2025-06-30

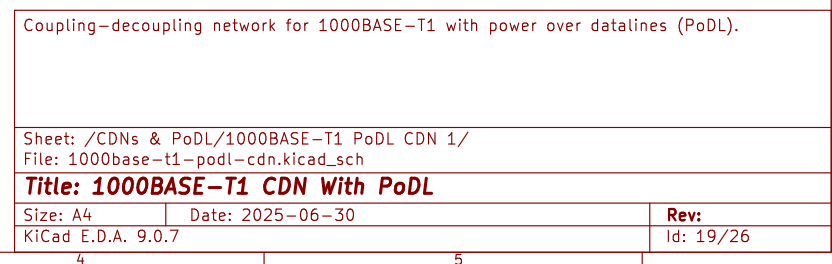
KiCad E.D.A. 9.0.7

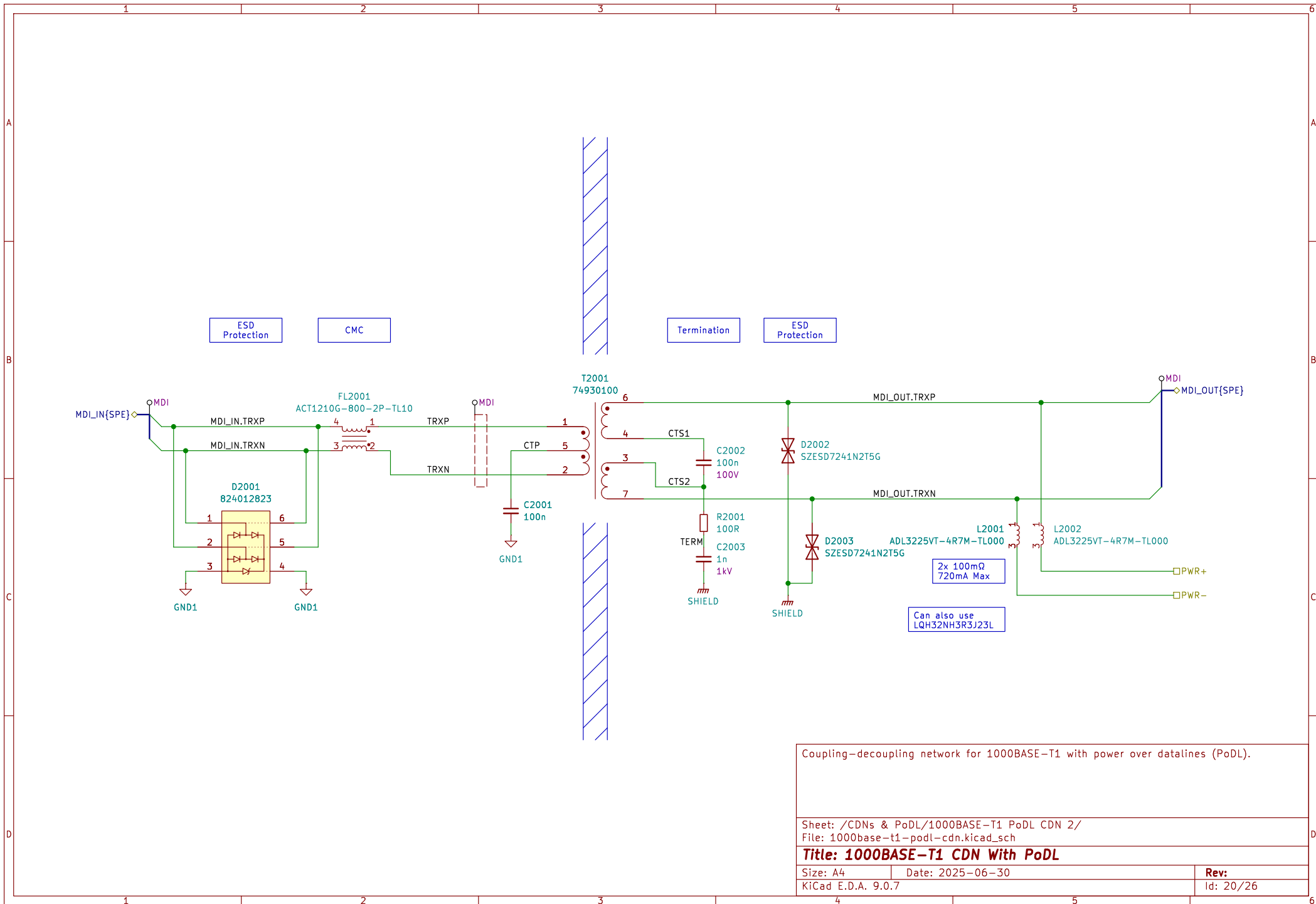
Rev:

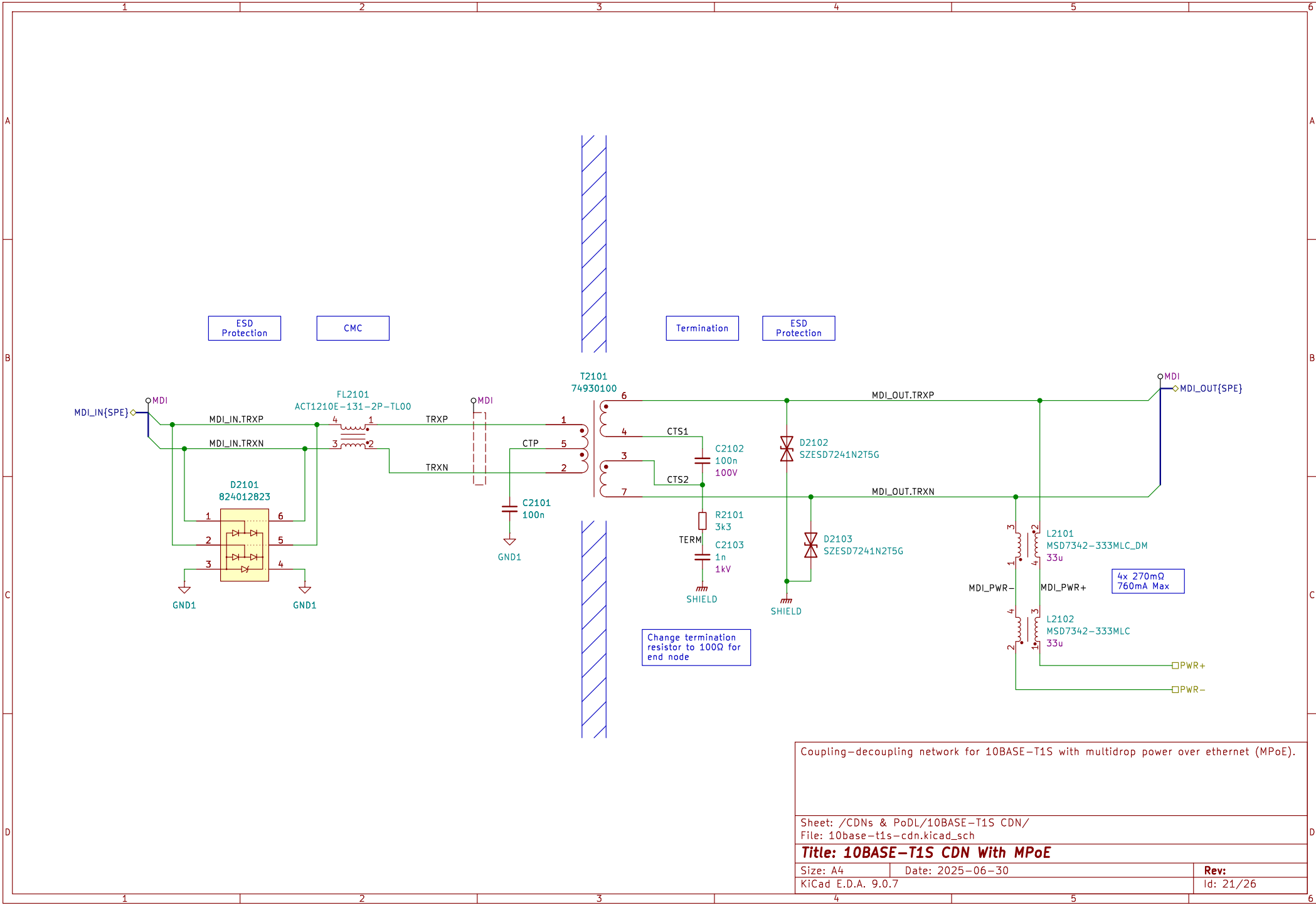
Id: 17/26



Coupling-decoupling network for 1000BASE-T1		
Sheet: /CDNs & PoDL/1000BASE-T1 CDN/		
File: 1000base-t1-cdn.kicad_sch		
Title: 1000BASE-T1 CDN		
Size: A4	Date: 2025-06-30	Rev:
KiCad E.D.A. 9.0.7	Id: 18/26	







Coupling-decoupling network for 10BASE-T1S with multidrop power over ethernet (MPoE).

Sheet: /CDNs & PoDL/10BASE-T1S CDN/
File: 10base-t1s-cdn.kicad_sch

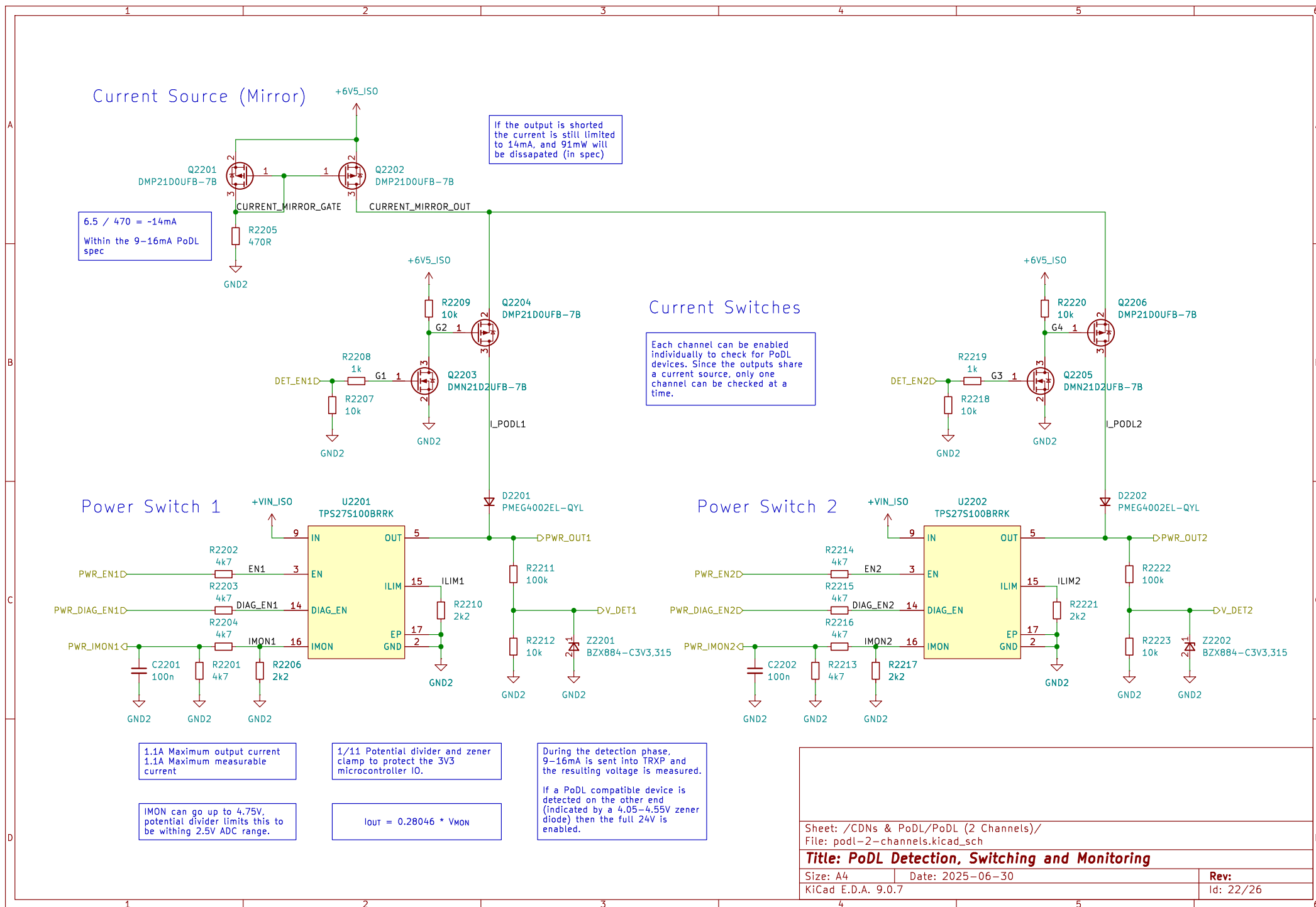
Title: 10BASE-T1S CDN With MPoE

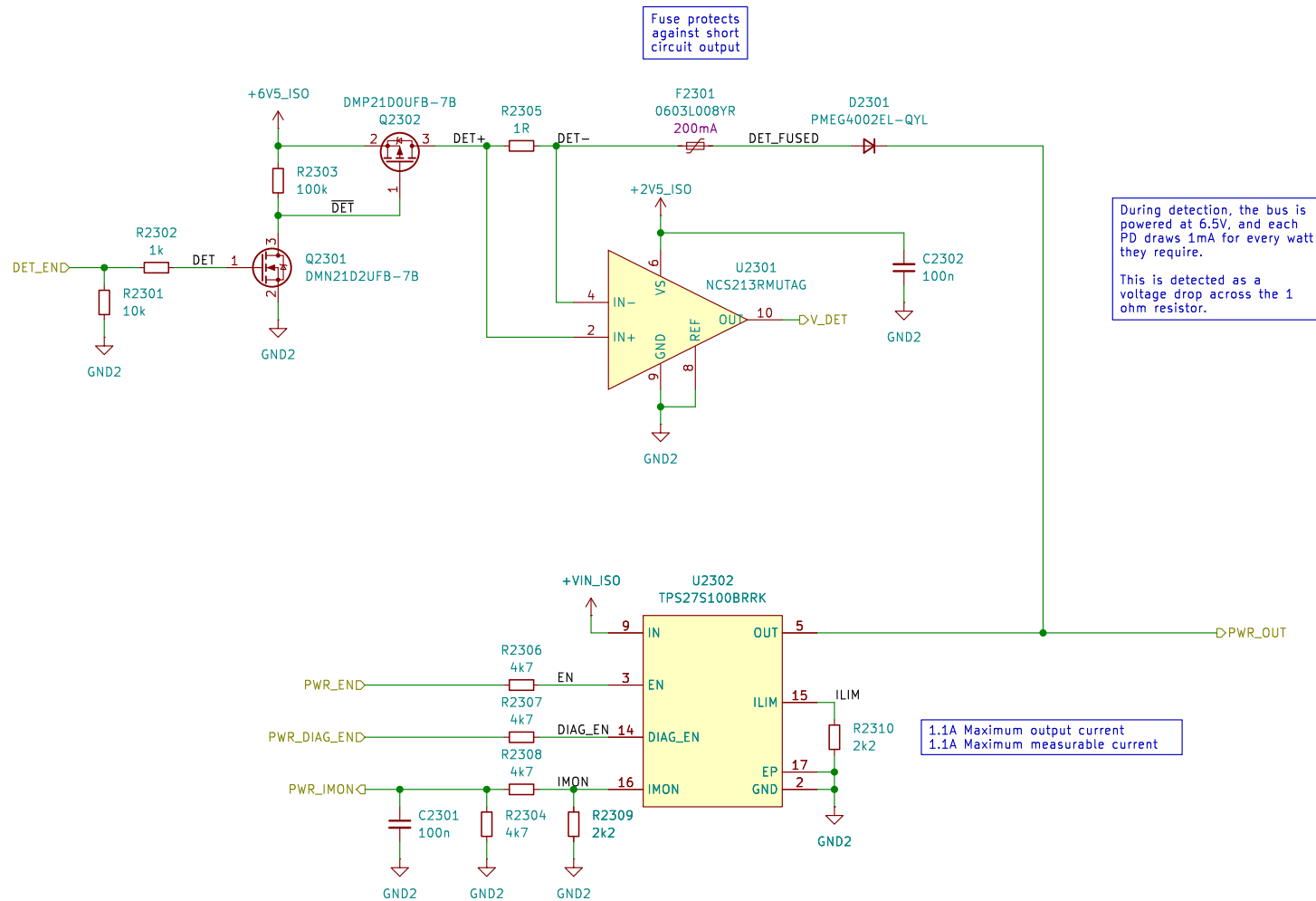
Size: A4 Date: 2025-06-30

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Sheet: /CDNs & PoDL/MPoE (1 Channel)/
File: mpoe-1-channel.kicad_sch

Title: MPoE Detection, Switching and Monitoring

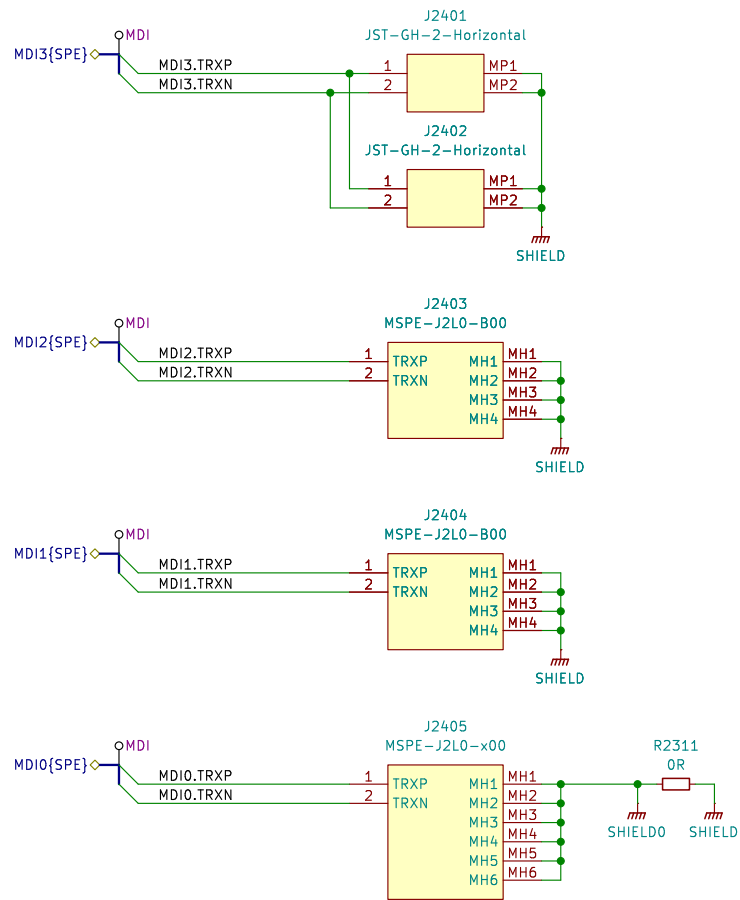
Size: A4

Date: 2025-06-30

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Sheet: /Ethernet Connectors/
File: ethernet-connectors.kicad_sch

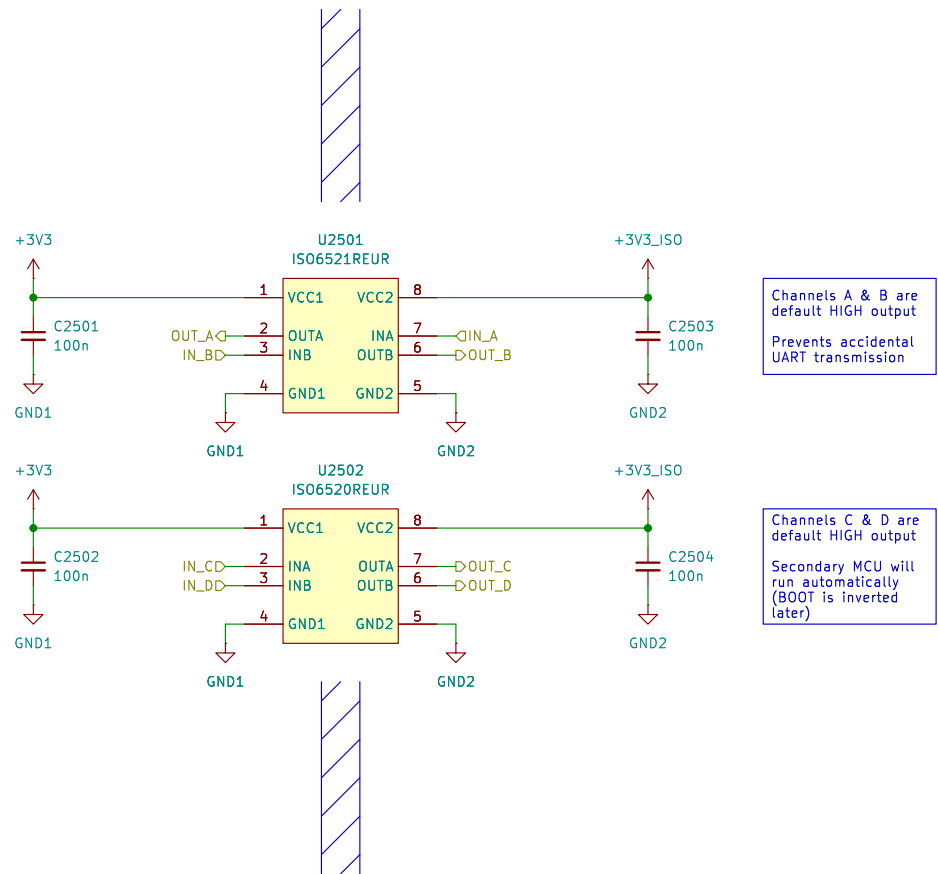
Title: Ethernet Connectors

Size: A4 Date: 2025-06-30

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Sheet: /Isolation Crossing/ File: isolation-crossing.kicad_sch		
Title: Isolation Crossing		
Size: A4	Date: 2025-06-30	Rev:
KiCad E.D.A. 9.0.7	Id: 25/26	

